

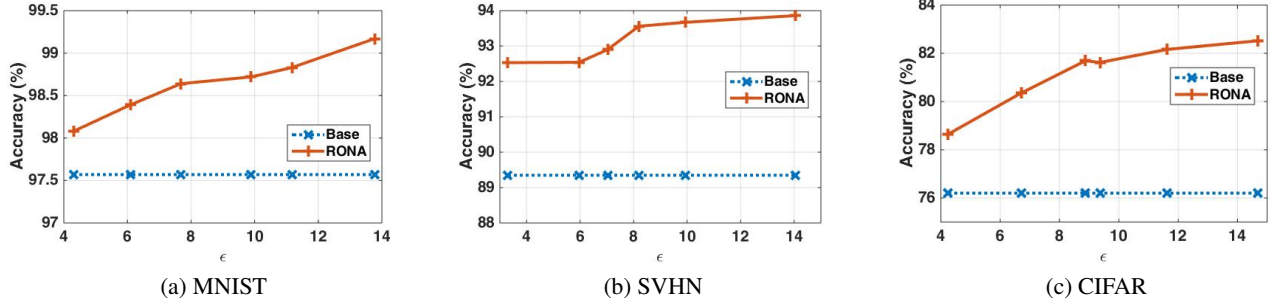


Figure 3: Accuracy vs. privacy budget ϵ . Base denotes the student model trained without querying the teacher model.

Table 1: Performance of Masking Specific Samples

	Base	RONA		
		$\epsilon = 5.2$	$\epsilon = 8.7$	$\epsilon = 29.8$
Overall	79.87	80.93	88.48	90.35
Unseen Classes	0.00	9.59	46.75	53.43

MNIST and SVHN with $(7.68, 10^{-5})$ and $(7.03, 10^{-6})$ differential privacy. It outperforms the results in (?) which achieved 98.10% and 90.66% accuracy on $(8.03, 10^{-5})$ guaranteed MNIST and $(8.19, 10^{-6})$ guaranteed SVHN. It is comparable with the latest results in (?), achieving 98.5% and 91.6% accuracy on $(1.97, 10^{-5})$ guaranteed MNIST and $(4.96, 10^{-6})$ guaranteed SVHN, which however used better and larger baseline networks. On CIFAR-10, we obtain 81.69% accuracy with $(8.87, 10^{-5})$ privacy, which outperforms 73% accuracy with $(8.00, 10^{-5})$ privacy in (?). RONA’s performance is even better than the latest cloud-based solution that achieved 79.52% accuracy on CIFAR-10 (?).

We verify the performance with different privacy budgets on three datasets. Fig. 3 demonstrates that the accuracies on three datasets generally rise with the increase of privacy budget. Thanks to the techniques proposed in this work, the student can still benefit from the knowledge distillation even when the knowledge is protected by a strong privacy.

In the above experiments, we select the public data randomly from the original training samples. We further test RONA in a much tougher condition on MNIST where we regard all training samples of the digits 6 and 9 as sensitive data. This data masking mimics a possible contingency in reality when some specific kinds of samples are highly sensitive. For the student model, 6 and 9 are mythical digits it has never seen. As listed in Table 1, without the teacher’s knowledge, the student cannot recognize 6 and 9, getting accuracy of 0. RONA can significantly improve the student’s accuracy on 6 and 9 with a reasonable privacy loss even though the student has never seen 6 and 9 during training.

Compression Performance

We randomly choose 80% training samples as the public data to validate the compression performance on the three datasets. For MNIST, SVHN, and CIFAR-10, we enforce

Table 2: Compression Performance

		# Params	Time(s)	Acc(%)
MNIST	T	155.21K	0.76	99.48
	S1	4.97K	0.03	98.94
	S2	9.88K	0.07	99.28
SVHN	T	1.41M	7.34	96.36
	S1	0.04M	0.29	94.49
	S2	0.07M	0.39	95.39
CIFAR-10	T	3.12M	13.92	86.35
	S1	0.15M	0.93	82.14
	S2	0.52M	3.10	84.57

$(9.60, 10^{-5})$, $(9.83, 10^{-6})$, and $(9.59, 10^{-5})$ differential privacy, respectively. In order to examine the runtime on mobile devices, we deploy these DNNs on HUAWEI HONOR 8 equipped with ARM Cortex-A53@2.3GHz and Cortex-A53@1.81GHz to process 100 images consecutively.

The results listed in Table 2 show that the models with larger sizes achieve better performance. On all the three datasets, the student models trained by RONA obtain comparable accuracies to the teacher models in spite of using much less capacity and fewer training data. On MNIST, the student achieves $15\times$ compression ratio and $11\times$ speed-up with only 0.2% accuracy decrease. On SVHN, the student model obtains slightly worse result ($< 1\%$) than the teacher model, while requiring 20 times fewer parameters and 19 times less runtime. On CIFAR-10, the accuracy decreases less than 2% while the model size is 6 times smaller. The above results argue that RONA can privately compress large models with acceptable accuracy loss.

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